INSTITUTE OF DISTANCE LEARNING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI, GHANA

AE 351
SOIL AND WATER ENGINEERING I

Lecture 1

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Introduction

The Soil and Water Engineering course:

- focuses on the soil/water interface and
- how to manage it for maximum productivity within the environmental requirements of long-term sustainability and overall ecosystem health.

The course brings together:

❖aspects of hydrology as applied in engineering,

❖soil mechanics and soil science, which are closely related to one another for the rational management of water resources for basic engineering structures targeted at mechanized agricultural development especially in irrigation and soil conservation.

COURSE OBJECTIVE

The general objective of the course is:

- to equip students with the requisite knowledge to be able to collect and analyse basic agro-hydrology and soil mechanics data to solve soil and water engineering problems.

COURSE CONTENT

- **UNIT 1:** Definition/Importance/Applications of Hydrology in Engineering and Agriculture
- **UNIT 2:** Analysis of Meteorological Data Such as Precipitation and Evapotranspiration
- **UNIT 3:** Infiltration, Percolation and Groundwater Hydrology
- **UNIT 4:** Surface Runoff and Hydrograph Analysis
- **UNIT 5:** Hydrological Forecasting and Design Criteria for Hydrological Structures
- **UNIT 6:** Basic Soil Mechanics: Concepts and Applications in Engineering and Agriculture

PRACTICAL SESSIONS

- Field Infiltration Tests
- Determination of Moisture Contents of Soils
- Determination of Bulk Density
- Collection and Analysis of Meteorological and Hydrological Data

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- Ofori, E. and Kyei-Baffour, N. (2010). Agrometeorology of some selected crops: Agrometeorology and Maize Production. Chapter 10.3 of **Guide to Agricultural Meteorological Practices** (GAMP). 3rd Edition No. 134 by the World Meteorological Organisation/Technical Committee for Agro-Meteorology (WMO/CAgM), 10: 25-35 & 110-111 [online] accessed on 15 October 2014 from http://www.wmo.int

UNIT 1: DEFINITION/IMPORTANCE/APPLICATIONS OF HYDROLOGY IN ENGINEERING AND AGRICULTURE

Hydrology is the science that studies:

- the occurrence,
- distribution,
- movement and properties of the waters of the earth and
- their relationship with the environment within each phase of the hydrological cycle.

Hydrology is dependent on climate, topography and geology.

Hydrology in Engineering

- Hydrologists apply scientific knowledge and mathematical principles to solve water-related problems in society.
- These are problems of quantity, quality and availability of water.
- The also make mathematical models of all hydrological phenomena.
- They may be concerned with finding water supplies for cities or irrigated farms, or controlling river flooding or soil erosion.
- They may work in environmental protection

Some Applications of Hydrology

- Hydrology is used to determine maximum probable flood at proposed sites e.g. Dams.
- The variation of water production from catchments can be calculated and described by hydrology.
- Determine the relationship between a catchment's surface water and groundwater resources
- The expected flood flows over a spillway, at a highway culvert, or in an urban storm drainage system can be known by this very subject.

Some Applications of Hydrology continued

- It helps us to determine the required reservoir capacity to assure adequate water for irrigation or municipal water supply in droughts condition.
- It tells us what hydrologic hardware (e.g. rain gauges, stream gauges, etc.) and software (computer models) are needed for real-time flood forecasting
- Used in connection with design and operations of hydraulic structure
- Used in prediction of flood over a spillway, at highway culvert or in urban storm drainage
- Used to assess the reservoir capacity required to assure adequate water for irrigation or municipal water supply during drought

Some Applications of Hydrology continued

- Hydrology is an indispensable tool in planning and building hydraulic structures.
- Hydrology is used for city water supply design which is based on catchments area, amount of rainfall, dry period, storage capacity, runoff evaporation and transpiration.
- Dam construction, reservoir capacity, spillway capacity, sizes of water supply pipelines and the effect of afforestation on water supply schemes, are all designed on basis of hydrological equations
- Hydrology is also used in solving problems in military planning and operations.

Uses of Hydrology in Engineering:

- Determining the water balance of a region
- Determining the agricultural water balance
- Designing riparian restoration projects
- Mitigating and predicting flood, landslide and drought risk

Uses of Hydrology in Engineering continued:

- Real-time flood forecasting and flood warning
- Designing irrigation schemes and managing agricultural productivity
- Part of the hazard module in catastrophe modelling
- Providing drinking water
- Designing dams for water supply or hydroelectric power generation

Uses of Hydrology in Engineering continued:

- Designing bridges
- Designing sewers and urban drainage systems
- Analyzing the impacts of antecedent moisture on sanitary sewer systems
- Predicting geomorphological changes, such as erosion or sedimentation
- Assessing the impacts of natural and anthropogenic environmental change on water resources and
- Assessing contaminant transport risk and establishing environmental policy guidelines.

Thank You

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